

PHY697 and PHY699

MSc Research Project II

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Program: PhD

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Tentative Title: Quasi-Normal Modes in $f(R)$ Gravity

Problem Statement: The goal of this project is to investigate the quasi-normal modes of black hole solutions in various $f(R)$ gravity models. The focus will be on identifying the differences in QNM spectra compared to GR and analyzing the physical implications of these deviations. Additionally, the project will explore the behavior of QNMs in the conformally related Einstein frame, where the scalar degree of freedom (scalaron) in gravity manifests as a scalar field. By studying both frames, we aim to understand the interplay between the metric perturbations and the scalar field perturbations and their role in the stability of black holes in $f(R)$ gravity theories.